

TONGULAH, Russian Federation

WMO: 248430

Lat: 61.917N

Lon: 124.550E

Elev: 653

StdP: 14.35

Time Zone: 9.00 (E09)

Period: 83-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -61.5	(c) -57.0	(d) -64.3	(e) 0.1	(f) -61.6	(g) -59.8	(h) 0.2	(i) -56.9	(j) 10.1	(k) -18.5	(l) 8.7	(m) -18.8	(n) 0.1	(o) 20	(p) 0.723

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	27.7	85.6	66.5	81.3	64.3	77.3	62.0	69.0	81.6	66.3	78.4	63.5	74.1	4.1	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
64.8	93.9	74.8	61.5	83.7	71.3	59.0	76.3	69.2	33.7	82.0	31.4	79.0	29.3	73.5	81.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
11.9	10.0	8.7	-63.9	90.8	5.4	2.8	-67.8	92.8	-71.0	94.5	-74.0	96.1	-78.0	98.2		
			-63.9	73.6	5.4	4.1	-67.8	76.6	-71.0	79.0	-74.0	81.3	-78.0	84.3		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	16.1	-33.4	-23.0	-1.2	20.3	41.7	56.8	60.5	54.6	39.0	18.0	-13.7	-28.7
	DBStd	35.58	14.95	13.30	14.58	12.37	8.56	7.52	7.54	6.88	8.01	12.03	15.07	15.91
	HDD50	13131	2586	2043	1586	890	280	22	8	33	339	992	1911	2441
	HDD65	17901	3051	2463	2051	1340	721	260	181	330	780	1457	2361	2906
	CDD50	767	0	0	0	0	24	227	333	174	9	0	0	0
	CDD65	63	0	0	0	0	0	15	42	6	0	0	0	0
	CDH74	1585	0	0	0	0	17	458	870	234	6	0	0	0
	CDH80	481	0	0	0	0	3	123	301	54	0	0	0	0
Wind	WSAvg	2.6	1.0	1.8	2.8	3.7	4.0	3.3	2.9	2.6	2.9	2.9	2.1	1.4
Precipitation	PrecAvg	15.7	0.6	0.5	0.5	0.7	1.5	1.9	2.6	2.5	1.9	1.3	1.0	0.7
	PrecMax	21.4	1.3	1.2	1.1	1.6	2.6	3.5	5.9	6.5	3.5	2.4	2.1	1.5
	PrecMin	11.8	0.3	0.1	0.1	0.4	0.5	0.7	0.7	0.6	0.5	0.5	0.3	0.2
	PrecStd	2.3	0.2	0.2	0.2	0.3	0.6	0.7	1.0	1.1	0.7	0.4	0.4	0.2
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	10.6	18.2	43.2	55.8	77.1	88.0	91.5	86.1	73.4	47.4	32.7	20.3
		MCWB	9.7	16.4	34.9	43.1	57.2	64.9	68.0	67.9	56.5	39.8	29.9	18.8
	2%	DB	2.6	11.4	34.8	49.1	70.2	83.6	87.4	79.8	65.4	41.3	24.6	12.4
		MCWB	1.9	10.4	28.9	39.0	52.8	63.6	68.7	64.0	53.1	37.6	23.2	11.4
	5%	DB	-3.0	5.7	29.4	44.7	65.5	79.4	83.3	75.4	60.2	37.6	17.7	3.3
		MCWB	-3.6	4.7	25.2	35.7	50.1	61.2	67.1	61.3	50.4	33.6	16.6	2.5
	10%	DB	-9.8	-0.6	24.6	40.9	59.6	75.4	78.6	71.1	55.3	34.5	10.8	-4.6
		MCWB	-10.2	-1.3	21.4	33.8	47.1	59.8	64.7	59.4	47.0	31.6	9.8	-5.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.6	16.4	35.7	43.6	57.5	70.6	74.7	69.7	58.0	41.2	29.9	18.9
		MCDB	10.6	18.0	42.9	55.1	76.3	79.8	83.9	81.7	68.6	44.8	32.5	20.5
	2%	WB	2.0	10.4	29.8	39.9	53.6	66.1	70.4	65.8	54.8	38.0	23.2	11.2
		MCDB	2.6	11.4	34.6	48.4	67.6	79.7	82.3	76.6	63.5	41.7	24.7	12.3
	5%	WB	-3.5	4.7	25.2	36.4	51.1	62.8	68.0	63.2	51.6	34.1	16.5	2.4
		MCDB	-2.8	5.6	29.0	43.4	63.6	76.7	81.1	72.5	59.5	36.8	17.8	3.0
	10%	WB	-10.1	-1.3	21.5	33.9	47.9	60.4	65.5	60.6	48.0	31.7	9.8	-5.3
		MCDB	-9.7	-0.6	24.6	40.3	58.8	72.5	77.0	69.6	53.8	34.4	10.7	-4.8
Mean Daily Temperature Range	5% DB	MDBR	18.4	26.4	32.9	29.0	22.8	26.6	27.7	27.7	24.4	17.2	19.5	16.8
		MCDBR	23.1	28.0	30.5	30.9	33.4	35.3	35.7	37.0	35.6	17.8	21.5	21.7
		MCWBR	22.5	26.8	25.2	21.9	18.7	17.6	19.1	21.7	24.0	13.2	20.5	21.6
	5% WB	MCDBR	23.1	27.8	30.2	26.4	30.4	31.2	31.3	30.7	32.8	15.8	21.4	22.6
		MCWBR	22.6	26.7	25.4	18.8	17.9	16.9	18.7	18.9	23.1	12.6	20.4	22.1
Clear-Sky Solar Irradiance	taub	0.222	0.262	0.296	0.345	0.398	0.414	0.549	0.503	0.352	0.284	0.222	0.192	
	taud	1.957	2.064	2.085	2.101	2.215	2.267	1.854	1.933	2.361	2.337	1.993	1.879	
	Ebn at Noon	142	219	255	264	259	257	212	210	237	216	140	92	
	Edh at Noon	12	23	33	40	39	38	56	48	26	18	11	7	
All-Sky Solar Radiation	RadAvg	116	436	992	1631	1706	1870	1686	1273	784	386	168	55	
	RadStd	8	19	33	59	125	130	102	85	88	39	14	2	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (2 neighbors)	+1.44	N/A	N/A	N/A	N/A	N/A	-488	-500	N/A	N/A

Nomenclature: See separate page